product (maker, model, type)
PC (model, speed, ram, hd, price)
laptop (model, speed, ram, hd, screen, price)
printer (model, color, type, price)

**1. Find hard-disk sizes that occur in two or more PC's. (Using RA without COUNT)**

$$R1(model1, hd1) \leftarrow \pi_{model,\, hd}(PC)$$
$$R2(model2, hd2) \leftarrow \pi_{model,\, hd}(PC)$$
$$Result \leftarrow \pi_{hd1}(R1 \bowtie_{model1 \neq model2 \wedge hd1 = hd2} R2))$$

**2. Find the manufactures of laptops with at least two different speeds. (Using RA without COUNT)**

$$R1(maker1, model1, speed1) \leftarrow \pi_{maker,\, model,\, hd}(product \bowtie laptop)$$
$$R2(maker2, model2, speed2) \leftarrow \pi_{maker,\, model,\, hd}(product \bowtie laptop)$$
$$Result \leftarrow \pi_{maker1}(R1 \bowtie_{maker1 = maker2 \wedge speed1 \neq speed2} R2))$$

**3. Find manufactures of PC's that may also make laptops. (Using TRC)**

$$\{\, p1.maker \mid p1 \in product\ ((p.maker = p1.maker) \wedge (p1.type = \text{“PC”}))$$
$$\wedge\ \exists p2 \in product\ ((p.maker = p2.maker) \wedge (p2.type = \text{“laptop”}))\ \}$$

**4. Find laptops that have a larger main memory than a PC and higher price than a PC. (Using DRC)**

$$\{a1 \mid \exists a3,a6\ (\langle a1, a2, a3, a4, a5, a6\rangle \in laptop$$
$$\wedge\ \exists b3,b5\ (\langle b1, b2, b3, b4, b5\rangle \in PC$$
$$\wedge\ (a3 > b3) \wedge (a6 > b5)))\}$$

**5. Find manufactures who sell PC's with the maximum price. (Using SQL without MAX)**

```
SELECT maker, price
FROM product JOIN PC ON (product.model = PC.model)
WHERE price >= ALL (
          SELECT price FROM PC
)

-- SQLite
SELECT maker, price
FROM product JOIN PC ON (product.model = PC.model)
```

```
WHERE price = (
          SELECT DISTINCT price
          FROM PC
          EXCEPT
          SELECT DISTINCT p1.price
          FROM PC p1 JOIN PC p2 ON p1.price < p2.price
)


-- SQLite
SELECT maker, price
FROM product JOIN PC ON (product.model = PC.model)
WHERE price = (
          SELECT DISTINCT p1.price
          FROM PC p1 LEFT JOIN PC p2 ON p1.price < p2.price
          WHERE p2.price is NULL
)
```

**6. Find the average price of PC's and laptops made by manufacturer "D". (Using SQL)**

```
SELECT
(SELECT
    (SELECT SUM(price)
    FROM product JOIN PC ON (product.model = PC.model)
            WHERE maker = 'D')
    +
    (SELECT SUM(price)
    FROM product JOIN laptop ON (product.model = laptop.model)
            WHERE maker = 'D'))

/

(SELECT
    (SELECT COUNT(*)
    FROM product JOIN PC ON (product.model = PC.model)
            WHERE maker = 'D')
    +
    (SELECT COUNT(*)
    FROM product JOIN laptop ON (product.model = laptop.model)
            WHERE maker = 'D'))
```

```sql
-- Using WITH
WITH maker_price AS (
     SELECT maker, price
     FROM product JOIN PC ON (product.model = PC.model)
     UNION ALL
     SELECT maker, price
     FROM product JOIN laptop ON (product.model = laptop.model)
)
SELECT maker, AVG(price)
FROM maker_price
WHERE maker = 'D'
```